

Compte rendu

PFSENSE

TP à rendre

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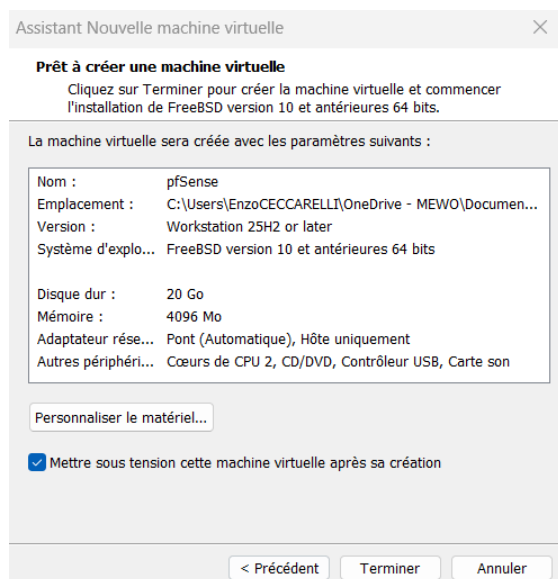
Introduction :

Dans ce TP, nous allons apprendre à installer PFSense. PFSense est un logiciel open source qui permet de transformer un ordinateur ou un serveur en pare-feu. Il permet de contrôler, sécuriser et gérer le trafic réseau.

Installation du logiciel :

Nécessaire :

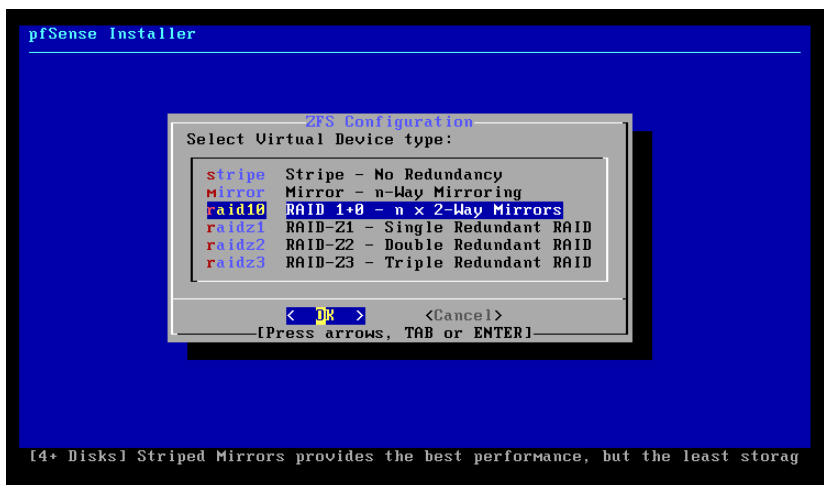
- VM ware
- ISO pfSense configuré (4Go de RAM / 2 Cœurs/2 cartes réseau : Une configuré par pont l'autre en privé à l'hôte).



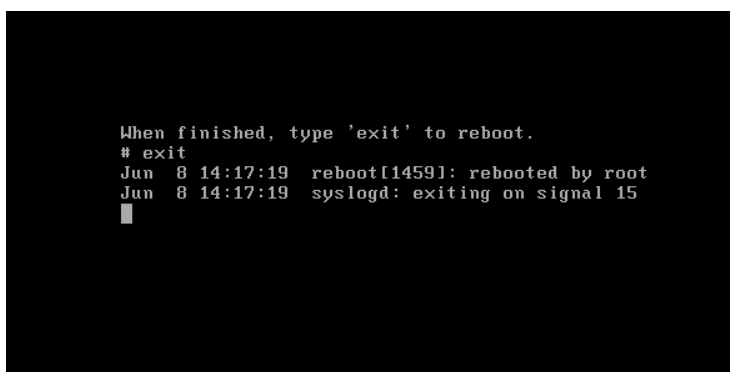
Lorsque vous lancer la vm pfSense va vous demander d'installer pfSense vous faites OK



Choisissez Stripe/espace pour cocher le disque dur puis ENTRER



Taper exit lors de cette étape



Une fois le téléchargement fini vous arriverez sur ce menu

```
done.
Starting CRON... done.
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 9d2806e04ddbb7117b6a

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 10.10.0.136/16
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)      9) pfTop
1) Assign Interfaces      10) Filter Logs
2) Set interface(s) IP address
3) Reset webConfigurator password
4) Reset to factory defaults
5) Reboot system
6) Halt system
7) Ping host
8) Shell
9) pfTop
10) Filter Logs
11) Restart webConfigurator
12) PHP shell + pfSense tools
13) Update from console
14) Enable Secure Shell (sshd)
15) Restore recent configuration
16) Restart PHP-FPM

Enter an option: █
```

Sur la capture d'écran présente si dessus, on peut constater que notre adresse ip via DHCP est bien présente (en bleu). C'est elle qui est connecté au réseau de l'école.

Tandis que l'adresse ip du réseau LAN (en orange) celui-ci est notre ip de l'adresse local sécurisé.

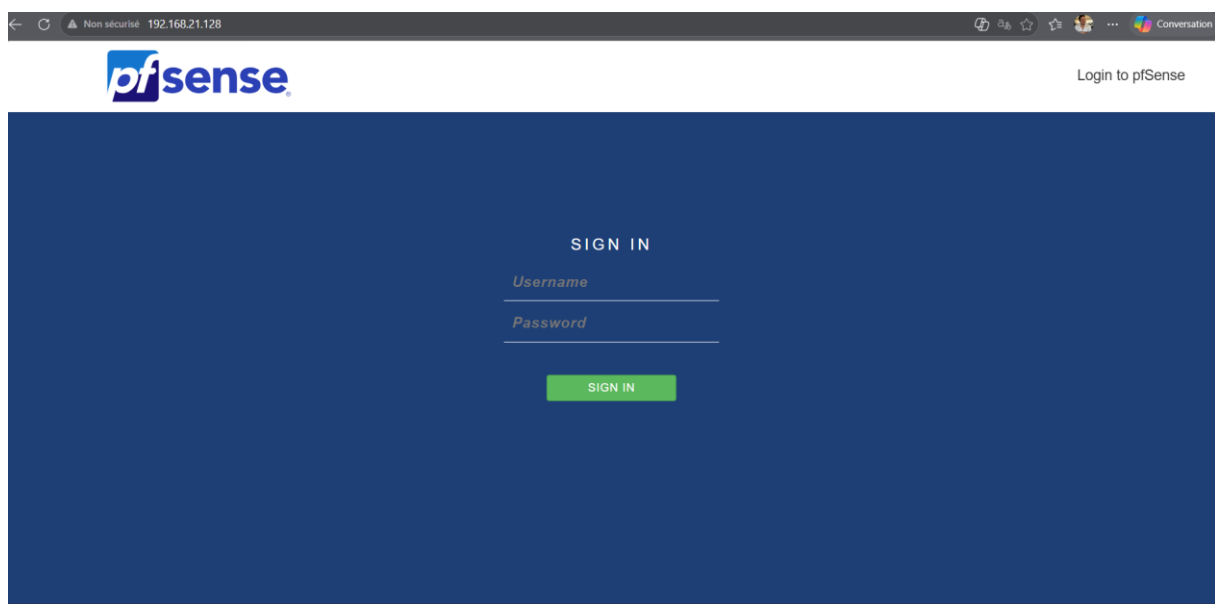
Notre mise en place est finie.

Configuration à l'interface graphique :

Pour cette étape nous allons ouvrir notre navigateur, pour ma part ce sera Microsoft Edge.

Taper dans l'adresse ip de votre LAN

(Dans mon cas j'ai dû modifier l'adresse ip en tapant 2 et 2 dans le menu de pfSense.)

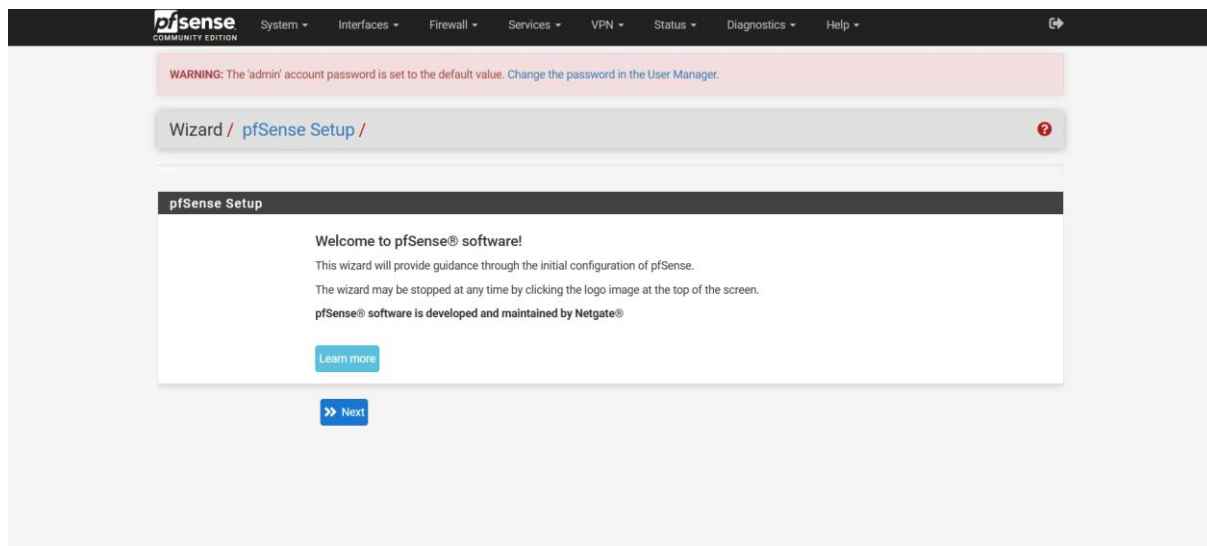


Pour se connecter il suffira d'utiliser le compte là :

Username : admin

Password : pfsense

Nous voici sur la page de pfSense.



Il nous suffira de faire « next » jusqu'à arriver sur cette page. Ici il faudra mettre **1.1.1.1** en DNS primaire et en second **8.8.8.8** pour avoir le DNS de Google.

On this screen the general pfSense parameters will be set.

Hostname	pfSense
Name of the firewall host, without domain part.	
Examples: pfsense, firewall, edgefw	
Domain	home.arpa
Domain name for the firewall.	
Examples: home.arpa, example.com	
Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.	
The default behavior of the DNS Resolver will ignore manually configured DNS servers for client queries and query root DNS servers directly. To use the manually configured DNS servers below for client queries, visit Services > DNS Resolver and enable DNS Query Forwarding after completing the wizard.	
Primary DNS Server	1.1.1.1
Secondary DNS Server	8.8.8.8
Override DNS	☒
Allow DNS servers to be overridden by DHCP/PPP on WAN	

Ici le changement se fera sur le Timezone en y mettant celui de Paris pour avoir la bonne heure

Wizard / pfSense Setup / Time Server Information ?

Step 3 of 9

Time Server Information

Please enter the time, date and time zone.

Time server hostname
Enter the hostname (FQDN) of the time server.

Timezone

[Next](#)

Descendre tout en bas de la prochaine case pour décocher la case Block RFC1918. Vu que c'est une VM pfSense va bloquer tout internet venant du réseau.

PPTP Local IP Address

pptplocalsubnet

PPTP Remote IP Address

PPTP Dial on demand ☐ Enable Dial-On-Demand mode
This option causes the interface to operate in dial-on-demand mode, allowing a virtual full time connection. The interface is configured, but the actual connection of the link is delayed until qualifying outgoing traffic is detected.

PPTP Idle timeout
If no qualifying outgoing packets are transmitted for the specified number of seconds, the connection is brought down. An idle timeout of zero disables this feature.

RFC1918 Networks

Block RFC1918 Private Networks ☐ Block private networks from entering via WAN
When set, this option blocks traffic from IP addresses that are reserved for private networks as per RFC 1918 (10/8, 172.16/12, 192.168/16) as well as loopback addresses (127/8). This option should generally be left turned on, unless the WAN network lies in such a private address space, too.

Block bogon networks

Block bogon networks ☒ Block non-Internet routed networks from entering via WAN
When set, this option blocks traffic from IP addresses that are reserved (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and obviously should not appear as the source address in any packets received.

[Next](#)

Lors de la page suivante il faut configurer notre interface de LAN en mettant notre adresse ip et le sous masque

Configure LAN Interface

On this screen the Local Area Network information will be configured.

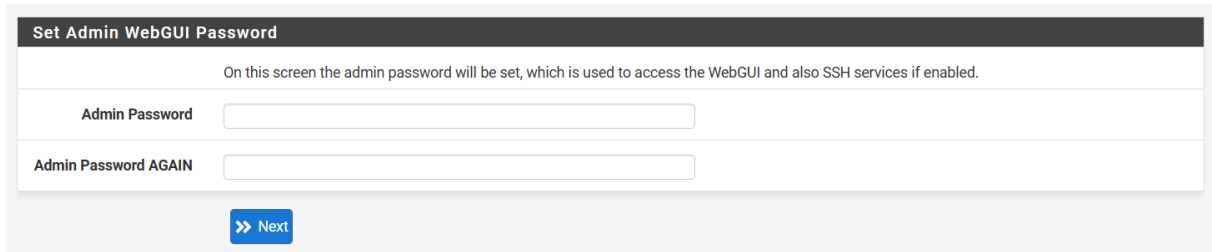
LAN IP Address
Type dhcp if this interface uses DHCP to obtain its IP address.

Subnet Mask

[Next](#)

A ce niveau, on va mettre un mot de passe sécuriser pour protéger notre réseau.

Le mot de passe de celui-ci sera *Azertyuiop123+*.



Les étapes sont fini ! il ne reste plus qu'à cliquer sur finish en bas de la page.

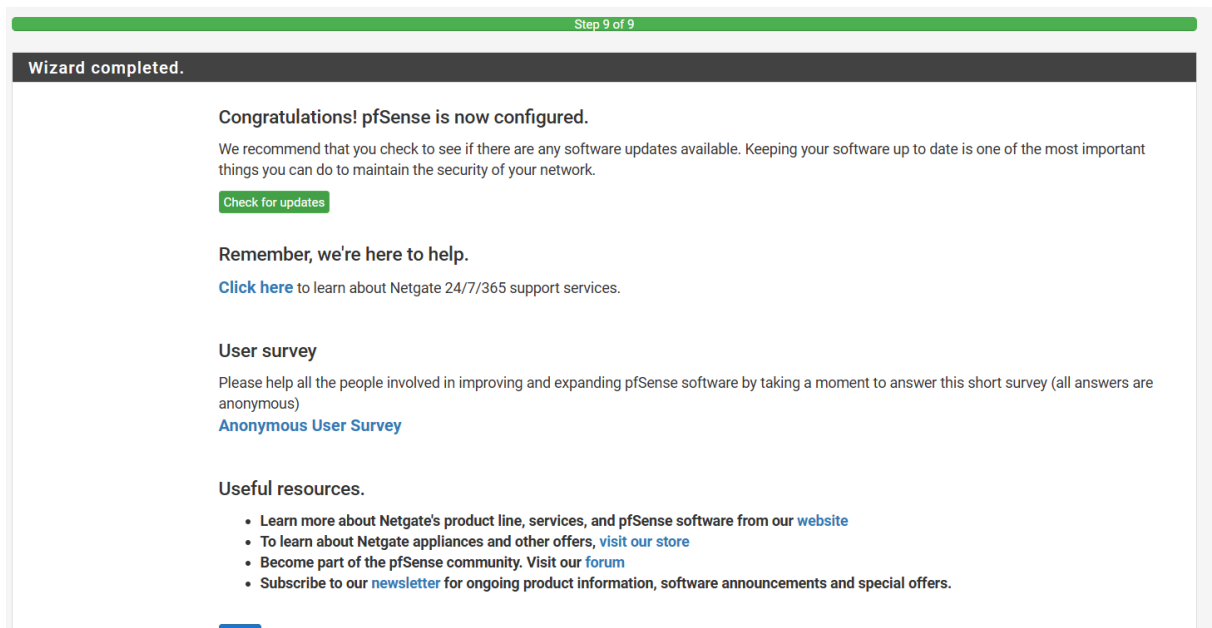


Tableau de bord de pfSense :

The screenshot shows the pfSense dashboard interface. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into two panels. The left panel, titled 'System Information', displays details about the pfSense installation, including the user (admin@192.168.21.1), system type (VMware Virtual Machine), BIOS version (6.00), and the current pfSense version (2.7.2-RELEASE). The right panel, titled 'Netgate Services And Support', provides information about the support contract type (Community Support) and lists various resources available to users, such as the Netgate Resource Library, upgrade options, and support subscriptions.

System Information	
Name	pfSense.home.arpa
User	admin@192.168.21.1 (Local Database)
System	VMware Virtual Machine Netgate Device ID: 9d2806e04ddb7117b6a
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Mon Mar 24 2025
Version	2.7.2-RELEASE (amd64) built on Wed Dec 6 21:10:00 CET 2023 FreeBSD 14.0-CURRENT The system is on the latest version. Version information updated at Mon Jun 8 15:17:40 CEST 2026
CPU Type	Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz 2 CPUs: 1 package(s) x 2 core(s) AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	Inactive
Kernel PTI	Enabled
MDS Mitigation	Inactive
Uptime	00 Hour 20 Minutes 34 Seconds

Netgate Services And Support	
Contract type	Community Support Community Support Only
NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES	
If you purchased your pfSense gateway firewall appliance from Netgate and elected Community Support at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the NETGATE RESOURCE LIBRARY. You also may upgrade to a Netgate Global Technical Assistance Center (TAC) Support subscription. We're always on! Our team is staffed 24x7x365 and committed to delivering enterprise-class, worldwide support at a price point that is more than competitive when compared to others in our space. - Upgrade Your Support - Netgate Global Support FAQ - Netgate Professional Services - Community Support Resources - Official pfSense Training by Netgate - Visit Netgate.com If you decide to purchase a Netgate Global TAC Support subscription, you MUST have your Netgate Device ID (NDI) from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase TAC supports here.	

Voici le tableau de bord sur lequel tout se passera lors des prochaines étapes, tel que l'installation d'une machine test ou encore découvrir les menus.

Conclusion :

Ce TP m'a permis de découvrir pfSense grâce au déploiement et à la mise en place d'un pare-feu. J'ai su accomplir les tâches demandées malgré certaines erreurs que j'avais pu faire, tel que remettre une nouvelle adresse IP à mon LAN. La partie la plus complexe était lors de la configuration du pare-feu car il fallait bien vérifier les éléments qui étaient présents sur ma VM pfSense et ceux que je devais mettre sur l'interface via l'adresse IP.

Le pare-feu est prêt à sécuriser notre réseau virtuel.